GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

Pure Math With Stats 3
0770/3

JUNE 2023 **ADVANCED LEVEL**

Subject Title	**Pure Mathematics With Statistics**
Paper No.	**Paper 3**
Subject Code No.	**0770**

THREE HOURS

Full marks may be obtained for answers to ALL questions.

Mathematical formulae and tables produced by the GCE Board are allowed.

In calculations, you are advised to show all the steps in your working, giving the answer at each stage.

Electronic calculators may be used.

Start each question on a fresh page.

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1. (i) Two events A and B are such that $P(A) = \frac{2}{3}$, $P(A \cup B) = \frac{3}{4}$, and $P(A \cap B) = \frac{5}{12}$.

Find,

(a) $P(B)$,

(b) $P(A/B)$,

(c) $P(B \cap A')$. **(3, 2, 3 marks)**

(ii) A manufacturer of phones employs an inspector to check the qualities of his product. After testing a large sample from a consignment, the inspector finds out that 2 out of every 10 phones are defective. A customer buys two of the phones.

Calculate, the probability that

(d) both phones are defective

(e) exactly one phone is defective. **(2, 3 marks)**

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2. The yields, to the nearest kilograms of 150 fruit trees, are given in the table below

Yield, x kg	10 - 14	15 - 19	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44	45 - 49
Number of trees, f	4	9	18	28	39	30	15	7

Estimate, to two decimal places,

(a) the mean yield,

(b) the standard deviation of the yields.

(c) Draw a cumulative frequency curve for the data.

Using the cumulative frequency curve, estimate

(d) the median yield

(e) the quartile deviation of the distribution. **(3, 3, 2, 2, 3 marks)**

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3. (i) A certain vaccine is administered to a group of persons. It is observed that 0.1% of all the persons who take the vaccine suffered from a side effect.

Using the Poisson distribution, find, to four decimal places, the probability that out of 1000 persons who take the vaccine,

(a) none will suffer from the side effect.

(b) exactly two persons will suffer from the side effect.

(c) more than two persons will suffer from the side effect. **(3, 2, 3 marks)**

(ii) Out of 1000 customers that visit a telephone booth, 60 buy credit cards.

Given that the number of customers buying credit follows a binomial distribution, find, the probability that in a randomly selected sample of 10 customers,

(d) exactly three buy credit cards

(e) more than seven buy credit cards **(2, 3 marks)**

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4. A discrete random variable X has probability mass function, f, defined by

$$f(x) = \begin{cases} k(6-x) \text{ for } x = 0, 1, 2, 3, 4, 5, \\ 0, \quad \text{elsewhere.} \end{cases}$$

Find,

(a) the value of the constant k.

(b) the mean of X.

(c) the variance of X.

Another random variable Y is such that $Y = 4X + 5$.

Find

(d) the mean of Y

(e) the variance of Y

(3, 3, 3, 2, 2 marks)

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5. A continuous random variable X has probability density function, f, defined by

$$f(x) = \begin{cases} \frac{1}{16}x(kx + 12), 0 \leq x \leq 2 \\ 0, \quad \text{elsewhere.} \end{cases}$$

Find (a) the value of the constant k.

(b) the mean of X.

(c) the variance of X.

Show that

(d) the median, m, of X satisfies the equation $m^3 - 6m^2 + 8 = 0$ **(3, 3, 3, 4 marks)**

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6. (i) State the conditions under which the normal distribution can be used to approximate the binomial distribution.

(ii) A coin is tossed 100 times.

Given that the random variable X is "the number of heads obtained",

(a) Find $E(X)$ and $\text{Var}(X)$.

Use the normal approximation to the binomial distribution to find, to four decimal places,

(b) $P(X < 45)$,

(c) $P(X > 60)$,

(d) $P(45 < X < 60)$. **(2, 3, 2, 3, 3 marks)**

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7. (i) A sample of n independent observations is taken from a normal population with mean 74 and variance 36.

(a) State the distribution of the sample mean, $\bar{X}$.

(b) Given that $P(\bar{X} > 75) = 0.282$, find the value of n. **(2, 3 marks)**

(ii) The masses of tubers of yam harvested from a certain farm are normally distributed with mean 145grams and standard deviation 9 grams. A random sample of 100 yams tubers selected from the farm is found to have a mean mass of 140 grams. Stating clearly, the null and alternative hypotheses, test, at the 5% level of significance whether the mean mass of the yam tubers is reduced. **(8 marks)**

8. The marks of eight students in English and Mathematics are as shown in the table below.

English , x	50	58	35	86	76	43	40	60
Mathematics, y	65	72	54	82	32	74	40	53

Calculate, to two decimal places,

(a) the Pearson product-moment correlation coefficient for these data.
(b) the Spearman's rank correlation coefficient for these data.
Find,
(c) the least squares regression lines of x on y and of y on x.
You may use $\sum xy = 26762$, $\sum x^2 = 27310$ and $\sum y^2 = 29958$. **(5, 4, 4 marks)**